SOCAL WILDFIRES GRID CO-SIMULATION

A validated extreme-event testbed for infrastructure resilience

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Onboarding for new project members

MOTIVATION

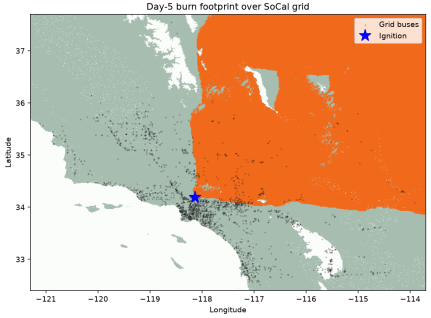
Why This Project Exists

January 2025, Los Angeles basin

- The **Eaton** and **Palisades** fires burned through the LA basin under Santa-Ana forcing.
- Wind-aligned fronts advanced at **60–80 m/min**, with spotting beyond 1 km.
- Distribution circuits, communication links and control systems were overwhelmed *simultaneously*.
- Consequence was not just burned area — it was **loss of electric service** to hundreds of thousands of customers.

The chain this project models

fire spread → asset damage → grid state → resilience metrics



Simulated fire perimeter, day 5

(`analysis/fire_perimeter_day5.png`)

The Research Gap

Fire models and grid models rarely talk to each other

Fire-centric	Grid-centric	This testbed
Detailed spread physics (Rothermel, Cell2Fire, FARSITE).	Detailed power-flow and contingency analysis.	Validated CA spread *and* full power-flow re-solve, every step, on **one shared geospatial frame**.
Infrastructure is *absent* or a static value layer.	Fire enters as an *exogenous* outage list or PSPS scenario.	A burned cell and its electrical asset are always co-registered.

- Mao et al. (2026) explicitly call for *feedback-coupled fire/grid simulations*.
- Without coupling you cannot ask: *did that suppression tactic protect the right cells?*

Source: Veith & Sultan, "When Wildfire Breaks the Grid", Sec. 1–2.

The problem

- DRL controllers are trained on data they have seen.
- Catastrophic events are, by definition, **rare** — they sit outside the training distribution.
- So agents fail *exactly* when resilience matters most. This is the **Curse of Rarity**.

The GUARDIAN answer

- Synthesise physically plausible worst-case worlds from curated GIS data.
- Train foundation policies against them via **Adversarial Resilience Learning (ARL)**.

Project facts

NATO Science for Peace and Security Multi-Year Project.

Pillars: energy security, environmental hazards, advanced AI.

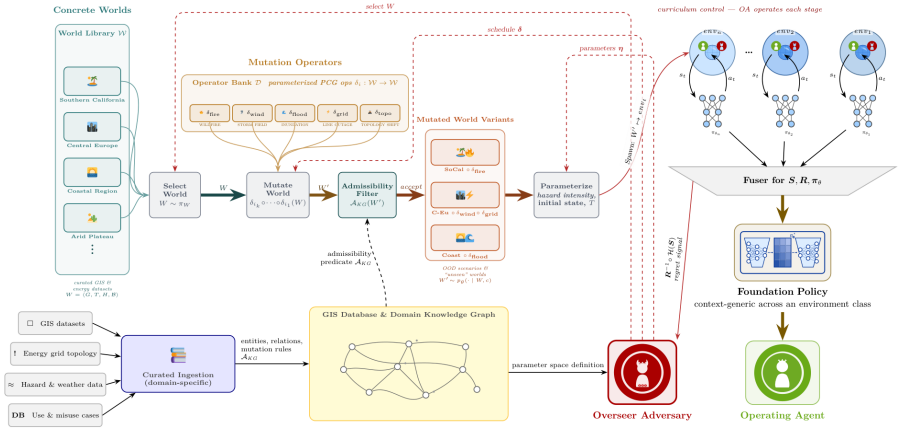
Dual PI: **Veith** (OFFIS, DE) & **Sultan** (Cal State LA, US).

Validation cases: SoCal wildfire cascades; Central European severe-weather grid failures.

Motivating precedent: the Feb. 2022 KA-SAT cyberattack disabled 5,800 wind turbines (≈11 GW) across Central Europe.

GUARDIAN System Architecture

Where this repository sits in the bigger picture



This repository implements the *SoCal* $\circ\, \delta_{fire}$ branch: one concrete world, one mutation operator, one operating agent.

Figure 2 from Sultan, Logemann & Veith, "GUARDIAN".

What You Will Take Away

Learning objectives

1. **Purpose** — what question this testbed is built to answer, and what it deliberately does *not* model.
2. **Concepts** — the constrained-mutation formalism CMA $= (S, \tau, D, \Theta)$ and where learning enters.
3. **Architecture** — how the geospatial, fire, grid and agent layers fit together in palaestrAI.
4. **Navigation** — which directory and which module to open for a given question.
5. **Operation** — the exact commands to reproduce every published figure.
6. **Research** — open ARL questions that are available as thesis topics.

Prerequisite

Basic palaestrAI familiarity. Slide 8 is a short refresher; a one-page reminder of experiment runs, environments and muscles.

PURPOSE AND CONCEPTUAL MODEL

What the Repository Delivers

Four deliverables

1 — NOAA-enriched MIDAS scenario

SoCal co-simulation scenario with weather switched from DWD Bremen to **NOAA HRRR** reanalysis, inside the scenario.

2 — palaestrAI environment

SoCal MIDAS wrapped as a first-class `palaestrai.environment.Environment`, plus an experiment run file.

3 — Wildfire agent (GUARDIAN CMA)

Cellular automaton driven by the Overseer-Adversary parameter vector Θ, over the full SoCal GIS footprint; optional PostGIS layer.

4 — Five-day co-simulation

120 h Santa-Ana wildfire/grid run with KPIs, plots and a generated report.

Source: repository `README.md`.

palaestrAI Refresher

The five nouns you need for this deck

Experiment run — A declarative YAML document: environments, agents, phases, termination conditions. Fully reproducible — the document *is* the experiment.

Environment — Owns state, publishes **sensors**, accepts **actuators**, is stepped by a simulation controller.

Agent — A **Brain** (learns, lives in the Learner process) plus one or more **Muscles** (act, live in RolloutWorkers).

Objective — Maps sensor readings to scalar reward. Separate from the environment, so several agents can score the same world differently.

Store — Every sensor, actuator and reward is written to a database, so analysis scripts read results instead of re-running simulations.

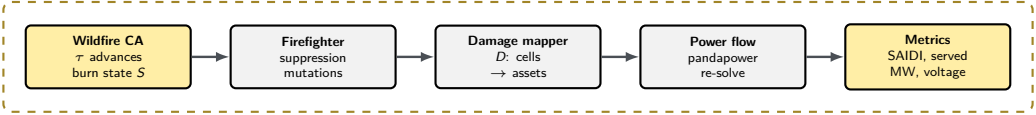
Two conventions used throughout

`TakingTurnsSimulationController` — agents act in a fixed order, not simultaneously.

`DummyBrain` — an agent that *acts but never learns*.

End-to-End Simulation Loop

One timestep



all four stages share one geospatial reference frame

- Agents act in **fixed turn order**: `wildfire` $\rightarrow$ `firefighter` $\rightarrow$ `damage_mapper`.
- Conflicting cell edits are resolved by *priority*, not by turn order — so the result is deterministic.

Mental Model: CMA $= (S, \tau, D, \Theta)$

The GUARDIAN constrained-mutation automaton

S — state

`int8` cell grid, co-registered with the raster. `UNBURNED` 0, `BURNING` 1, `BURNED_OUT` 2, `SUPPRESSED` 3, `CONTAINED` 5.

τ — transition

The CA sub-step (`dt_cma_min`, default 5 min). Each burning cell attempts to ignite its 8 Moore neighbours

D — damage map

Fails buses whose cell is burning or burned, and overhead lines inside a radiant-heat clearance buffer.

Θ — the adversary's handle

Ignition point(s), wind vector, dead-fuel moisture, and the global ROS multiplier κ ($1 \leq \kappa \leq 8$).

Θ **is the entire attack surface** — everything else is physics.

Source: `docs/CMA_AGENT.md`, `wildfire_cma/cma.py`.

What Is — and Is Not — Learning

The single most common misreading of this repo

Component	Brain	Learns?
Wildfire (GUARDIAN CMA)	`DummyBrain`	**No** — constrained mutation operator driven by Θ
Damage mapper	`DummyBrain`	**No** — deterministic consequence model
Firefighter v0.3–v0.5	scripted doctrine	**No** — expert incident-command heuristics
Firefighter v0.7	`FirefighterSacBrain`	**Yes** — SAC with CQL, bootstrapped offline

Why the fire does not learn

This is a deliberate, paper-faithful choice. The published work positions the CA as *the calibrated, validated substrate* on which a learned scenario-generation loop can *later* be built.

Why that matters to you

Turning Θ from a calibrated constant into a learned adversarial policy is the open door into ARL — and the subject of Sec. 6 of this talk.

TECHNICAL ARCHITECTURE

Repository Map

Where to look for what

Directory	Contents
`wildfire_cma/`	The CA itself: `cma.py` (S, τ, Θ), `gis.py`, `damage.py`, `wind_field.py`, `postgis.py`. Pure `numpy`, no palaestrAI import.
`palaestrai_socal/`	Environments (`gis_world_env.py`, `midas_grid_env.py`), all agents under `agents/`, and every `experiment*.yml`.
`socal_grid/`	Geo-referenced pandapower model plus `dispatch_and_run.py` (the convergence recipe) and `build_network.py`.
`midas_socal/`	MIDAS scenario `socal_midas.yml`, NOAA weather provider, `prepare_midas.py`, `run_sim.sh`.
`analysis/`	Drivers, report generators, timelapse builders and the committed result figures.
`data/`	GIS layers, DEM fetcher, CAL FIRE perimeters, offline teacher datasets, CAISO load.
`docs/, tests/`	Design docs; ≈30 test modules (**235 passing** as of v0.7).

Start reading at `wildfire_cma/cma.py` — it has no framework dependencies.

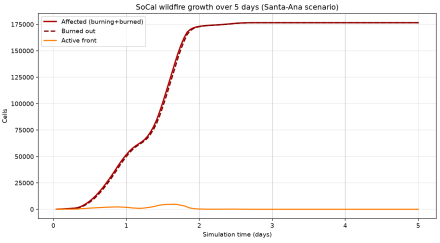
Geospatial Layer

The shared reference frame

- Footprint `SOCAL_BOUNDS` $= (-121.3,\ 32.4,\ -113.7,\ 37.7)$, EPSG:4326. Row 0 is the *north* edge.
- Three raster sources, in order of fidelity:
  - `from_rasters()` — real LANDFIRE FBFM40 fuel + 3DEP elevation (needs `rasterio`).
  - `socal_from_srtm()` — SRTM GL3 (≈90 m) via OpenTopography, 4×3 tile mosaic, resampled to a 600×760 model grid.
  - `synthetic_socal()` — deterministic fallback used by CI.
- Degrades *gracefully*: a missing DEM cache silently falls back to synthetic terrain.

Scale trap

On the statewide grid one cell ≈ 947 m. Retardant drops are sub-cell there — which is why fine grids (≈50 m) are used for suppression studies.



Fire growth on the statewide raster

(`analysis/fire_growth.png`)

Wildfire Spread Kernel

Rothermel-derived, wind- and slope-modulated

Rate of spread

$$R(i,j \to i',j') \;=\; \kappa \cdot R^{0}(i,j) \cdot \varphi_w(u,\theta) \cdot \varphi_s(z,\theta)$$

Ignition probability per sub-step

$$p \;=\; 1 - \exp(-\,R \cdot \Delta t_{\mathrm{CMA}}/\delta)$$

- R^0 — base ROS by fuel class, damped by dead-fuel moisture through the Rothermel coefficient η_m; zero above `FUEL_MX_EXTINCTION`.
- $\varphi_w = \exp(c\,u\cos\alpha)$, $c = 0.25$, tuned so $\approx$20 m/s Santa-Ana wind reproduces the observed **60–80 m/min** fronts.
- $\varphi_s = 1 + 5.275\tan^2(\text{slope})$· upslope alignment.
- δ = cell size in m ($\times\sqrt{2}$ diagonally). A neighbour ignites iff `rng.random() < p` — the only stochastic element, and it is seeded.

#	Fuel class	R^0 m/min
0	non-burnable	0.0
1	grass (GR)	4.0
2	grass-shrub (GS)	2.5
3	shrub / chaparral (SH)	**3.2**
4	timber-understory (TU)	1.2
5	timber-litter (TL)	0.8
6	slash-blowdown (SB)	1.5

Class 3 dominates SoCal. `gis.fbfm40_to_class()` maps the 40 Scott-Burgan models onto these seven.

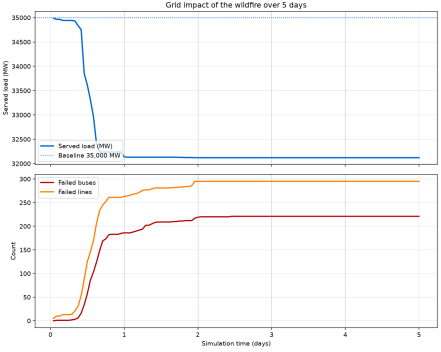
Power Grid Model

The electrical side

- **2,294 buses / 2,595 lines**, geo-referenced pandapower model (`socal_grid/socal_grid.json`, EPSG:4326, ≈5.9 MB).
- Built from open data: OSM substations, transmission lines, power plants, city populations, CAISO load profiles.
- An *illustrative* distribution topology — deliberately **not** a utility feeder model.
- Customers derived via `CUSTOMERS_PER_MW = 200.0`.

Convergence gotcha — read this first

Raw `pp.runpp()` on `socal_grid.json` **diverges**. Always go through `socal_grid/dispatch_and_run.py` — "config D": strengthen the network, co-locate dispatch, then solve with **Iwamoto Newton-Raphson**.



Grid impact over the five-day run

(`analysis/grid_impact.png`)

Damage Mapping

From burned cells to de-energised assets

The mechanism

1. Every bus and every line endpoint is projected onto the raster once, at setup.
2. A bus whose cell is `BURNING` or `BURNED_OUT` is de-energised.
3. An overhead line is removed if its corridor intersects a burning cell within the *radiant-heat clearance buffer*.
4. Power flow is re-solved; SAIDI and served MW are recorded.

Implementation note

mosaik exposes no `in_service` actuator for buses or lines. So under MIDAS the damage is applied as a **load-shed trip**: write `...load-<bus>-<idx>.p_mw` $\rightarrow 0$.

Mutation arbitration

Several agents may edit the same cell in one step. A fixed priority makes the outcome independent of turn order:

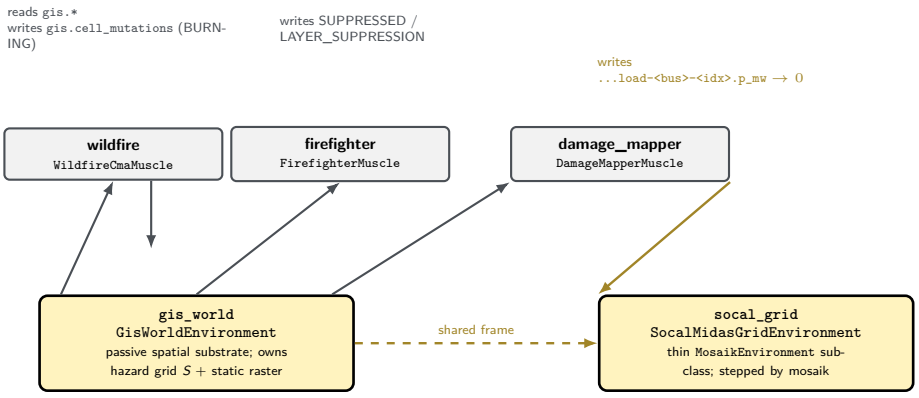
`BURNED_OUT` (terminal) $>$ `SUPPRESSED` $>$ `FLOODED` $>$ `BURNING` $>$ `UNBURNED`

See `spaces.arbitrate_mutations`. Suppression is not permanent: the environment reverts a cell after `SUPPRESS_PERSIST_STEPS` $= 12$.

Source: `docs/CMA_AGENT.md`, `docs/AGENTS.md`.

Multi-Agent Architecture

The v0.2 two-environment design



- `gis_world` is **hazard-agnostic**: it steps no dynamics, it only applies `gis.cell_mutations` and re-publishes telemetry. Wildfire today, flood or storm tomorrow.
- Ignition is an *agent parameter* (`ignition_points`, `ignition_step`), **not** an environment actuator.

Firefighting Model Evolution

Five releases, increasing realism

Tunable knobs (v0.4 onward)

`n_planes`, `n_helos`, `n_crews`, `n_dozers`, `n_engines`, `doctrine` $\in$ {`auto`, `direct`, `indirect`}, `protect_assets`.

Physical constraints are enforced

Fleets degrade above `DEGRADE_WIND_MS` $= 13$ m/s and are **grounded** at `GROUND_WIND_MS` $= 18$ m/s — so a 25 m/s high-wind Eaton run is genuinely unchanged by air support.

Source: `CHANGELOG.md`, `agents/firefighter_core.py`.

EXISTING RESULTS

Calibration Against Real Fires

The credibility argument

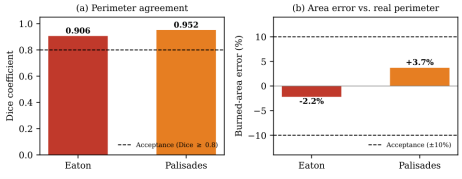


Figure 1 from "When Wildfire Breaks the Grid".

Fire	Dice	Area err.	Verdict
Eaton	0.906	$-2.2\,\%$	PASS
Palisades	0.952	$+3.7\,\%$	PASS

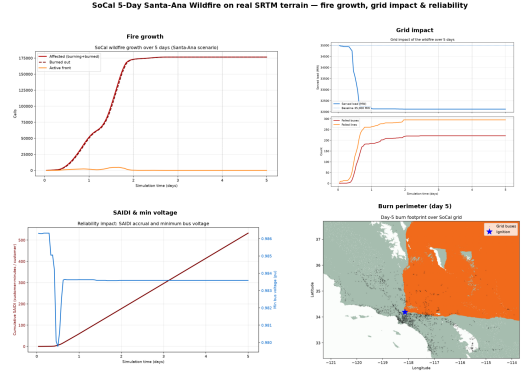
- Acceptance bar: Dice ≥ 0.8 **and** $|\text{area}\,\%| \leq 10$. Both fires clear it.
- Reference: official CAL FIRE perimeters at env step 60. Eaton 13,786 vs 14,102 ac; Palisades 24,677 vs 23,799 ac.

Honest caveat

Without containment the free-running CA *overshoots* by $+145\,\%$ (Eaton) and $+129\,\%$ (Palisades). The match relies on `wind_field.py`: perimeter-informed wind, fuel reclassification and `contain_burnable_footprint(..., margin_cells=2)`.

Five-Day Santa-Ana Baseline

The headline scenario



analysis/five_day_combined.png

Setup — 120 hourly steps, seed 47, synthetic 600×760 raster. Strong dry offshore wind on days 1–3, marine-layer recovery on days 4–5.

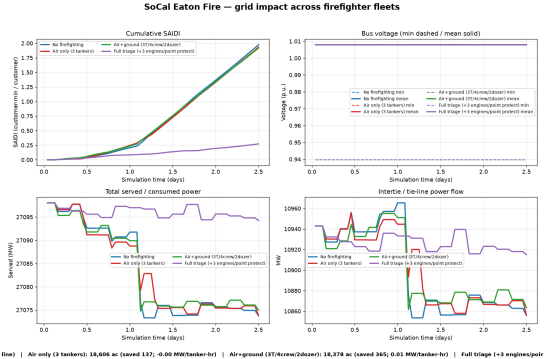
Outcome

- Served load ≈35,000 MW → **31,328 MW**
- ≈213 buses and ≈327 lines de-energised
- Peak ≈**734,000 customers** disconnected, around day 2.5
- Final burn footprint ≈166,000 cells
- Power flow converged **120/120** steps

Reproduce with `python analysis/run_5day.py --max-steps 120`.

Suppression Counterfactuals

Four phases of incident response



analysis/grid_metrics_v04.png

Phase	Acres saved	SAIDI
phase_0_no_ff	—	≈1.97
phase_1_air	137	≈1.95
phase_2_air_ground	365	≈1.93
phase_3_full_triage	**654**	**0.27**

The key insight of the whole project

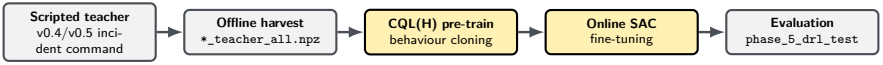
Full triage cuts total burned area by only a few percent — yet it drops SAIDI from ≈1.97 to **0.27** and keeps ≈20 MW more load energised.

A tactic that barely moves total burned area can still be decisive for grid resilience — if it protects the right cells.

On the Eaton fine grid, full triage holds SAIDI 1.61 → 0.24 (≈5,588 ac saved); Palisades ≈1,811 ac saved.

The Learning Firefighter

v0.7 — current frontier



The learning contract

Box(17) observation — burning/burned/suppressed fractions, fire-front geometry, mean slope, wind trigonometry, served MW, SAIDI delta, step progress. *Not* a raster flatten.

Discrete(4) action — `NOOP`, `INDIRECT`, `DIRECT`, `TRIAGE`. The policy picks a *doctrine*; the same IncidentCommand machinery executes it, gated by resource availability.

Reward — `SaidiObjective`: $r = -\Delta\text{SAIDI}/\text{scale}$, always ≤ 0, exactly 0 when load is fully served.

Why gate the actions?

So the learner *cannot invent impossible suppression*. It chooses tactics, not physics.

State this honestly

The v0.7 verification was a **short pipeline smoke run on synthetic-terrain fallback** — not a real-DEM result. Those SAIDI numbers are *not* metric-comparable to production, and two episodes cannot train a policy.

Production long-runs are configured (`episodes: 400`, `cql_alpha: 1.0`, `use_real_dem: true`) but not yet reported.

Two Bugs Worth Knowing About

Because you will hit their cousins

1 — Ragged sensor arrays

Symptom. palaestrAI's `Memory` crashed with `ValueError: All arrays must be of the same length`.

Cause. `gis.cell_state` (≈23,660 elements) was mixed with scalar `*-load-*.p_mw` sensors in one DataFrame.

Fix. `agents/_memory_compat.py`, a ragged-safe, idempotent shim for `_MuscleMemory._infos_to_df` — installed in *both* the RolloutWorker *and* the Learner process.

2 — The silent no-op learner

Symptom. hARL logged "could not update"; the firefighter trained for hours *without ever learning*.

Cause. Action-shape mismatch. The muscle emitted `np.array([act_id])` with shape `(1,)`, while the offline loader stored a 0-d scalar. `SACBrain.update()` builds a single `np.array(actions)`, which then went ragged.

Lesson. A DRL agent that silently fails to update looks exactly like a DRL agent that has not converged yet. Assert on your shapes.

Source: `CHANGELOG.md`, v0.7.

RUNNING THE TESTBED

Quick Start

Fifteen minutes to your first figure

1 — Install and verify

```
pip install -r requirements.txt      # or requirements-dev.txt
pytest -m unit                       # < 5 s, numpy only
pytest -m slow                       # loads the 5.9 MB grid, runs power flow
```

2 — Reproduce the headline result (no broker, no database)

```
python analysis/run_5day.py --max-steps 120 --outdir analysis
python analysis/make_timelapse.py     # -> wildfire_timelapse.gif + .mp4
```

3 — Optional: real terrain (free OpenTopography key)

```
export OPENTOPOGRAPHY_API_KEY=...
python data/dem/fetch_dem_tiles.py    # -> socal_srtm_gl3.npz, ~71 MB, git-ignored
```

`run_5day.py` is a standalone driver. It writes `FIVE_DAY_ANALYSIS.md`, four PNGs and `five_day_kpis.csv` — and needs no palaestrAI runtime at all.

The Full palaestrAI Path

When you need agents, phases and a store

Standard run

```
python midas_socal/prepare_midas.py

palaestrai database-create

# validate the YAML before burning an hour
python -c "from palaestrai.experiment \
  import ExperimentRun; \
  ExperimentRun.load(
    'palaestrai_socal/experiment.yml')"

palaestrai start palaestrai_socal/experiment.yml
```

MIDAS scenario only

```
midas_socal/run_sim.sh            # full
midas_socal/run_sim.sh --smoke    # CI / quick
```

Multi-phase A/B comparison

```
env PYTHONPATH=$PWD palaestrai \
  -c runtime_pg_eaton.conf.yaml start \
  palaestrai_socal/experiment_eaton_local_ab.yml
```

`--phases` grammar

Comma-separated list; each item is `phase_uid`, `phase_uid:n_planes`, or `uid:n_planes:Label`.

No commas inside a label. Legacy `--phase-a`/`--phase-b` still work.

Pick a `runtime*.conf.yaml` to match your horizon: `runtime_short`, `runtime_full16`, `runtime_full120`.

Key Scripts and What They Produce

Your analysis toolbox

Script	Purpose	Output
`analysis/run_5day.py`	Standalone 120-step baseline run	report + 4 PNGs + KPI CSV
`analysis/verify_calibration.py`	Assert Dice and area error against CAL FIRE perimeters	pass/fail verdict
`analysis/perimeter_validation.py`	Compare simulated vs. official perimeter geometry	Dice, area, overlay
`analysis/grid_metrics_report.py`	SAIDI, voltage, served power across phases	`grid_metrics_*.png`
`analysis/make_timelapse.py`	Animate a single run over the Esri satellite basemap	GIF + MP4
`analysis/make_comparison_timelapse.py`	Side-by-side animation of several phases	GIF + MP4
`analysis/make_sim_vs_real.py`	Simulated vs. real perimeter figure	comparison PNG
`analysis/firefighter_report.py`	Scripted-suppression KPI summary	tables + figures
`analysis/drl_firefighter_report.py`	Learned vs. scripted responder	comparison report

Scripts that read a store take `--store <uri>` and `--phases`; standalone drivers do not. Basemap tiles are cached in `data/basemap_cache/` — no API key needed. Attribution: Esri, Maxar, Earthstar Geographics.

Reproducibility and Provenance

Read before you upgrade anything

Pinned for a reason

palaestrAI constrains `numpy<2` and `pandas==2.1.4`. Do **not** bump either without re-running the calibration verification — the CA is validated against a specific numerical stack.

Determinism

The only stochastic element is the seeded neighbour-ignition draw. Same seed and same Θ give the same fire, every time.

CI pipeline (`.gitlab-ci.yml`)

`lint`	flake8, on push
`unit`	`pytest -m unit`, on push
`system`	`pytest -m slow`, manual
`simulate`	NOAA smoke + 5-day, manual

`simulate` publishes its figures as artifacts.

Open data only

USGS 3DEP / SRTM GL3, LANDFIRE FBFM40, OSM, CAISO, NOAA HRRR, CAL FIRE perimeters. GPL-3.0-or-later.

ARL RESEARCH QUESTIONS

From Calibrated Substrate to Adversarial Learning

Where the open work begins

Today

Θ is a **calibrated constant**, set once from the validation values and held fixed. The wildfire agent uses `DummyBrain`: it acts, it never learns.

The published paper says this explicitly — the testbed is *"the calibrated, validated substrate on which a future learned-scenario-generation loop could be built."*

The ARL target

Replace the constant with a **learned Overseer-Adversary** that proposes Θ to maximise regret against the operating agent, subject to an *admissibility predicate* $\mathcal{A}_{KG}$ that keeps every proposed world physically plausible.

Both sides then improve together — an autocurriculum in the sense of Baker et al. (2020) and unsupervised environment design.

Why this repository is the right starting point

The hard part — a fire model whose output is *trustworthy* (Dice ≥ 0.9 against two real perimeters) and coupled to a solvable grid — is already done and test-guarded. What is missing is the learning loop around Θ.

Research Questions

Available as thesis topics

RQ1 — Adversarial scenario generation

How can Θ be *learned* rather than calibrated? Which admissibility constraints keep generated worlds physically plausible while still being genuinely hard?

RQ2 — Objective design for resilience

Which signal best represents grid resilience: SAIDI, served MW, fraction of customers connected, or time-to-recovery? How do learned policies differ under each, in the resilience-trapezoid metric space?

RQ3 — Out-of-distribution generalisation

Can a responder trained on Eaton transfer to Palisades, to synthetic fires, or to a different hazard entirely — and how do we measure that honestly?

RQ4 — Reward hacking and simulator artifacts

The suppression action space is deliberately gated. Where else could a policy exploit the model rather than the phenomenon, and how would we detect it?

Research Questions

Continued

RQ5 — The mutation-operator boundary

Where should the line sit between a *constrained mutation operator* (cheap, interpretable, provably admissible) and a *learned adversary* (expressive, but capable of drifting into the implausible)?

RQ6 — Catastrophic forgetting in multi-phase crises

A real crisis is sequential: fire, then outage, then restoration. How do we keep an agent competent at phase 1 while it learns phase 3? (Kirkpatrick et al., 2017.)

RQ7 — Cross-hazard reuse

`gis_world` is hazard-agnostic by construction. What does it take to add δ_{wind}, δ_{flood} or δ_{grid} from the GUARDIAN operator bank — and do policies trained on fire transfer at all?

A good first paper

Take RQ2. It needs no new physics — only a new `Objective` subclass and a careful comparison. The store already holds everything you need.

Summary and Your First Contribution

What you now have

- A fire model **validated** against two real 2025 perimeters (Dice 0.906 / 0.952).
- Bidirectional **fire-to-grid coupling** with a full power-flow re-solve each step.
- Four scripted suppression tiers as **counterfactuals**, plus a first learning responder.
- **235 passing tests**, open data only, GPL-3.0-or-later.
- A clear, documented **ARL extension path**.

Pick one, this week

1. **Reproduce.** Run `run_5day.py` and match the published figures.
2. **Read.** Open `wildfire_cma/cma.py` and find equations 6 and 7 in the code.
3. **Perturb.** Change one component of Θ and explain the effect on SAIDI.
4. **Propose.** Write half a page on one of RQ1–RQ7.

Repository

`gitlab.com/arl-experiments/socal-wildfires`

References and Further Reading

- V. Sultan, T. Logemann, E. M. S. P. Veith. *GUARDIAN: Geospatial Unseen-event Adversarial Reinforcement for Defense and Infrastructure Adaptation.*
- E. M. S. P. Veith, V. Sultan. *When Wildfire Breaks the Grid: A Validated Extreme-Event Testbed for Infrastructure Resilience.*
- R. C. Rothermel (1972). *A mathematical model for predicting fire spread in wildland fuels.* USDA Forest Service.
- C. Pais et al. (2021). *Cell2Fire: A cell-based forest fire growth model.*
- M. Panteli et al. (2017). *Power system resilience assessment* — the resilience trapezoid.
- M. Bruneau et al. (2003). *A framework to quantitatively assess and enhance seismic resilience of communities.*
- J. Kirkpatrick et al. (2017). *Overcoming catastrophic forgetting in neural networks.*
- B. Baker et al. (2020). *Emergent tool use from multi-agent autocurricula.*
- E. M. S. P. Veith et al. (2023). *palaestrAI* — the experiment platform underlying this testbed.
- IEEE Std 1366 (2012). *Guide for electric power distribution reliability indices* — the SAIDI definition used throughout.

QUESTIONS?

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